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Quantitative Aptitude

Formula – Guide

Set Theory & Venn Diagram

Set Theory

Power Set

- If a set A has n element, then the number of elements in the power set is 2^n.
- The number of relations from a set A to a set B = $2^{n(A)\times n(B)}$

Proper Subsets

- If a set A has n element, then the total number of subsets (including the set itself and the empty set) is 2^n.
- The number of proper subsets of A is $(2^n - 1)$ because the proper subset excludes the set itself from the total number of subsets. where n is the number of elements in set A.

Difference of Sets

- Difference between two sets A and B is

  $A - B = \{x : x \in A \text{ and } x \notin B\}$
- Symmetric Difference between two sets A and B is

  $A \Delta B = (A - B) \cup (B - A)$

Union (A ∪ B)

$A \cup B = \{x : x \in A \text{ or } x \in B\}$

Some properties of Union operation

- A ∪ B = B ∪ A
- (A ∪ B) ∪ C = A ∪ (B ∪ C)
- A ∪ φ = A
- A ∪ A = A
- U ∪ A = U

Intersection (A∩B)

$A \cap B = \{x : x \in A \text{ and } x \in B\}$

Some properties of intersection operation

- A ∩ B = B ∩ A
- (A ∩ B) ∩ C = A ∩ (B ∩ C)
- φ ∩ A = φ ; U ∩ A = A
- A ∩ A = A
- A ∩ (B ∪ C) = (A ∩ B) ∪ (A ∩ C)
- A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C)

Important results

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cup B) = n(A) + n(B)$ {when A and B are disjoint sets}
- $n(U) = n(A) + n(B) - n(A \cap B) + n((A \cup B)')$
- $n(A \cup B) = n(A - B) + n(B - A) + n(A \cap B)$
- $n(A - B) = n(A \cap B) - n(B)$
- $n(A - B) = n(A) - n(A \cap B)$
- $n(A^c) = n(A') = n(U) - n(A)$
- $n(P \cup Q \cup R) = n(P) + n(Q) + n(R) - n(P \cap Q) - n(Q \cap R) - n(R \cap P) + n(P \cap Q \cap R)$
- $A + A' = U$

De Morgan's Laws

- (A ∪ B)′ = A′ ∩ B′
- (A ∩ B)′ = A′ ∪ B′

Properties of Sets

- **Commutative property**
  - ➔ $A \cup B = B \cup A$
  - ➔ $A \cap B = B \cap A$
- **Associative Property**
  - ➔ $(A \cap B) \cap C = A \cap (B \cap C)$
  - ➔ $(A \cup B) \cup C = A \cup (B \cup C)$
- **Distributive Property**
  - ➔ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
  - ➔ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- **Identity Property**
  - ➔ $A \cup \varnothing = A$
  - ➔ $A \cap U = A$
- **Complement Property**
  - ➔ $A \cup A' = U$
- **Idempotent Property**
  - ➔ $A \cap A = A$
  - ➔ $A \cup A = A$